

Seeding In Cell Flats

FILLING TRAYS

When seeding in cell flats, uniform growth depends on evenly filling the trays. This task should therefore be done attentively. It's also crucial to use soil mix recipes designed specifically for growing seedlings in cell flats.

We use 2 different commercial potting mixes; one that is formulated for starting the seedlings and another for transplanting. The latter contains 15% compost, whereas the first one contains virtually none. Those mixes are certified organic and contain no synthetic wetting agents. Their salinity levels are also guaranteed, which takes out the need to measure before each batch.

- Before filling the trays, mix potting soil with water - we do this in a large plastic container. Adding water will cause peat to swell, thereby making it more porous. For an 85 litre bag, 3 litres of water is added. Mix everything with a shovel until the moisture is evenly distributed. For the sake of efficiency, we always make two batches (two bags of soil) at a time.
- To fill the trays, empty 4 litres of soil on top of each tray, scattering the substrate as uniformly as possible. For efficiency place 4 empty trays on your working table, filling them one after another. Err on the side of filling the trays with too much soil rather than not enough, breaking up the clumps at the same time.
- Then, lift up the tray approximately 6" (15 cm) and let it drop. Brush off the excess and ensure that all cells, especially those in the corners, are evenly filled.
- The trays are then piled one on top of one another, thereby creating a depression where the seed will be placed. The one on top of pile will be done last.
- The trays are now ready to be seeded.

SEEDING TRAYS

Flats are stacked on in front of your working area in piles of 4 or 5. The seeder is on your left side. Start off by gently sweeping the plates with a paint brush. This will remove any soil mix and help with the adherence of the seeds. Everything is now in place.

To begin, sprinkle the seeds on tops of the seeding plate. Not putting too many at once is the best way to prevent small seeds from crowding in the seeding holes. Turn on the vacuum and open the on/off valve.

Adjust airflow traps according to seed dimension. Small, light seeds require very little air flow and lower vacuum pressure (i.e.: bypass valve more open). Conversely, larger, heavier seeds require greater air-flow and higher vacuum pressure in order to defy gravity (bypass valve more closed).

To distribute seed evenly, tilt the seeder forward and backward, shaking it gently from side to side. Try to keep the seeds moving at all times. Next, gently tap the seeder on the seed collector tray to remove all excess seeds with your fingers.

Place the seeder over your tray, open the on/off valve to release the seeds and give it a little tap.

Brush off the plate (potting mix often sticks to the plate) and start again.

Seeded trays are stacked in a criss-cross position so that seeds won't stick on to the trays. Make piles of 4 or 5 trays.

The seeded trays are then brought onto the rolling benches and are lightly topped with dry potting soil mix to prevent them from drying out too quickly.

Trays should be arranged on the benches according to the matching numbers of cells (the 128 with the 128, the 72 with the 72, and so on). This way each cell will receive an equal amount of water when watering.

To increase fertility and inoculate the young seedlings with beneficial microorganisms, we spray them weekly with seaweed or compost tea.

WATERING TRAYS

Watering in the nursery is always done twice: the first time is to humidify the soil and the second is for the water to fully penetrate the cell. The total amount of water sprayed will depend on many different factors; including outside ambient temperatures and the amount of sunshine.

Watering must occur more frequently when sunny and inversely, hardly any watering is required when cloudy.

Watering should take place before 2 pm in the spring and 4 pm in the summer months.

The watering schedule can be modified according to growth stage of the seedling:

STAGE 1: Germination: Spray nozzle 1,000 micro hole

- Frequent watering, especially in sunny conditions
- Avoid humidity variations
- Keep surface damp
- Do not let dry

STAGE 2: Young seedlings Spray nozzle 400 micro holes

- Water quickly and as needed;
- The seedling will benefit from drying up on the surface, encouraging the radicle to penetrate the soil.

STAGE 3: Transplant Spray nozzle 400 micro holes

- Watering less frequent, but more abundant
- Allow soil to dry up between waterings
- Prepare seedling for outdoor conditions
- Avoid the proliferation of algae and mold.